PROJECT REPORT

ONLINE SHOPPING CHATBOT USING GOOGLE CLOUD DIALOGFLOW ES

A Cloud-Based Conversational Shopping Assistant

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Technology: Google Cloud Platform (GCP)

Primary Service: Dialogflow ES

Project Type: Cloud / Conversational AI Project

Academic Year: 2026

1. Abstract

The Online Shopping Chatbot is a conversational application developed using Google Cloud Dialogflow ES. The purpose of the project is to provide customers with a simple text-based shopping assistant that can understand common shopping requests and respond automatically. The chatbot can handle product searches, understand product categories, collect a customer's budget, add products to a cart, and accept order-status queries. Dialogflow intents are used to identify the customer's requirement, while entities are used to extract useful information such as product type and budget. The project demonstrates how cloud-based conversational AI can be used to improve customer interaction in an online shopping environment.

2. Introduction

Online shopping websites contain many products and features, so customers may need assistance while searching for products or checking orders. A chatbot can provide immediate assistance through natural-language conversation. This project uses Google Cloud Dialogflow ES to create a basic shopping chatbot without requiring a traditional rule-based application interface.

The project was designed as a beginner-friendly cloud project. The initial implementation focuses on Dialogflow intents, entities, training phrases, and responses. It can later be extended with Firestore and a serverless backend to provide real product, cart, and order data.

3. Problem Statement

Customers may spend time navigating an online shopping website to find products, understand available options, or obtain order information. The objective is to develop a conversational assistant that understands simple customer requests and provides relevant responses quickly.

4. Objectives

- To develop an online shopping chatbot using Google Cloud Dialogflow ES.
- To understand the working of intents, entities, training phrases, and responses.
- To allow customers to search for common product categories using natural language.
- To understand and capture a customer's shopping budget.
- To provide basic cart and order-status interactions.
- To gain practical experience with a cloud-based conversational AI service.
- To create a foundation that can later be integrated with databases and serverless services.

5. Technologies and Services Used

Technology / Service	Purpose	Role in Project
Google Cloud Platform	Cloud platform	Hosts and manages cloud resources
Dialogflow ES	Conversational AI	Understands user messages and generates chatbot responses
Intents	Request classification	Identifies what the customer wants to do
Entities	Information extraction	Identifies product type and other values
@sys.number	Number recognition	Recognizes budget values such as 50000

Web / Dialogflow test console	Testing	Used to test chatbot conversations

6. System Architecture

The implemented basic architecture is:

Customer

↓

Dialogflow ES Agent

↓

Intent Recognition + Entity Extraction

↓

Configured Responses

Future enhanced architecture can connect Dialogflow to a serverless backend and Firestore:

Customer → Dialogflow ES → Cloud Functions / Cloud Run → Firestore

Firestore → Products / Cart / Orders → Backend → Dialogflow response

7. Working of the Chatbot

User Input: The customer enters a natural-language request such as 'I want to buy a laptop'.

Intent Detection: Dialogflow compares the request with training phrases and identifies the appropriate intent.

Entity Extraction: Important values such as product category or budget are extracted from the message.

Response: Dialogflow sends the response configured for that intent.

Conversation: The user can continue with another request such as providing a budget or asking for order status.

8. Implementation Procedure

1. Create or select a Google Cloud project named Online-Shopping-Chatbot.

2. Open the Dialogflow ES console and create an agent named OnlineShoppingBot.

3. Set English as the default language and Asia/Kolkata as the time zone.

4. Create the SearchProduct intent with sample training phrases such as 'I want to buy a laptop' and 'Show me laptops'.

5. Create the ProductType entity with values such as Laptop, Mobile, Headphones, Camera, and Tablet.

6. Create the ProductSearch intent and annotate product names using the ProductType entity.

7. Configure the chatbot to ask the user for a budget.

8. Use the @sys.number system entity to recognize numeric budget values.

9. Create the AddToCart intent with phrases such as 'Add this product to my cart'.

10. Create the OrderStatus intent with phrases such as 'Where is my order?' and 'Track my order'.

11. Add suitable responses for each intent.

12. Use the Dialogflow Try it now panel to test the conversations.

9. Intents Implemented

Intent	Purpose	Example Training Phrase	Response / Action
SearchProduct	Recognize shopping requests	I want to buy a laptop	Ask what product the customer wants
ProductSearch	Search by product category	Show me laptops	Confirm product category and ask for budget
AddToCart	Handle cart requests	Add the laptop to my cart	Confirm that the product was added
OrderStatus	Handle order tracking	Where is my order?	Ask for the order ID

10. Entities Implemented

Entity	Example Values	Purpose
ProductType	Laptop, Mobile, Headphones, Camera, Tablet	Identifies the requested product category
@sys.number / Budget	30000, 50000, 70000	Captures the customer's budget

11. Sample Conversation

Customer	I want to buy a laptop.
Bot	Sure! I can help you find a product. What product are you looking for?
Customer	Show me laptops.
Bot	Great! I found products in that category. What is your budget?
Customer	My budget is 50000.
Bot	The budget value has been recognized and can be used for product filtering in an enhanced version.
Customer	Add one laptop to my cart.
Bot	Done! The product has been added to your cart. Would you like to place the order?
Customer	Where is my order?
Bot	Sure! Please provide your order ID.

12. Testing

Test Case	Input	Expected Component	Result
Search product	I want to buy a laptop	SearchProduct	Intent recognized
Product category	Show me laptops	ProductSearch	ProductType recognized
Budget	My budget is 50000	@sys.number	Number recognized
Add to cart	Add the laptop to my cart	AddToCart	Cart response displayed
Order tracking	Where is my order?	OrderStatus	Order ID requested

13. Advantages

- Provides a simple conversational shopping interface.
- Reduces the need for users to navigate multiple pages for basic queries.
- Uses natural-language input instead of fixed menu options.
- Dialogflow provides intent and entity recognition capabilities.
- The project can be extended with other Google Cloud services.
- It is suitable as a beginner cloud and conversational-AI project.

14. Limitations

- The basic version does not contain a live product database.
- Product prices and stock are not dynamically retrieved.
- Cart and order information are not persistent in the basic Dialogflow-only version.
- Payment processing is not implemented.
- Advanced natural-language cases may require additional training phrases.
- A production application would require authentication, secure backend APIs, monitoring, and additional testing.

15. Future Scope

- Connect Dialogflow with Firestore to store products, carts, and orders.
- Use Cloud Functions or Cloud Run for backend business logic.
- Implement real-time product search, price filtering, and stock checking.
- Add customer authentication and personalized order history.
- Integrate a secure payment gateway in a production-ready system.
- Create a web-based chat interface and deploy it publicly.
- Add multilingual support and voice interaction.
- Add Cloud Logging and monitoring for operational visibility.

16. Security Considerations

- Use least-privilege IAM permissions for cloud resources.
- Do not expose service-account private keys or API credentials.
- Validate user input before sending it to backend services.
- Protect customer and order information using appropriate access controls.
- For production deployment, use HTTPS and secure authentication.

- Monitor cloud resources and review logs for unexpected activity.

17. Conclusion

The Online Shopping Chatbot successfully demonstrates the basic use of Google Cloud Dialogflow ES for conversational applications. The project shows how intents can identify customer requests and entities can extract useful information such as product type and budget. The chatbot provides a foundation for an online shopping assistant and can be expanded using Firestore, Cloud Functions or Cloud Run, authentication, and a web interface. The project also provides practical exposure to cloud computing and conversational AI concepts.

18. Learning Outcomes

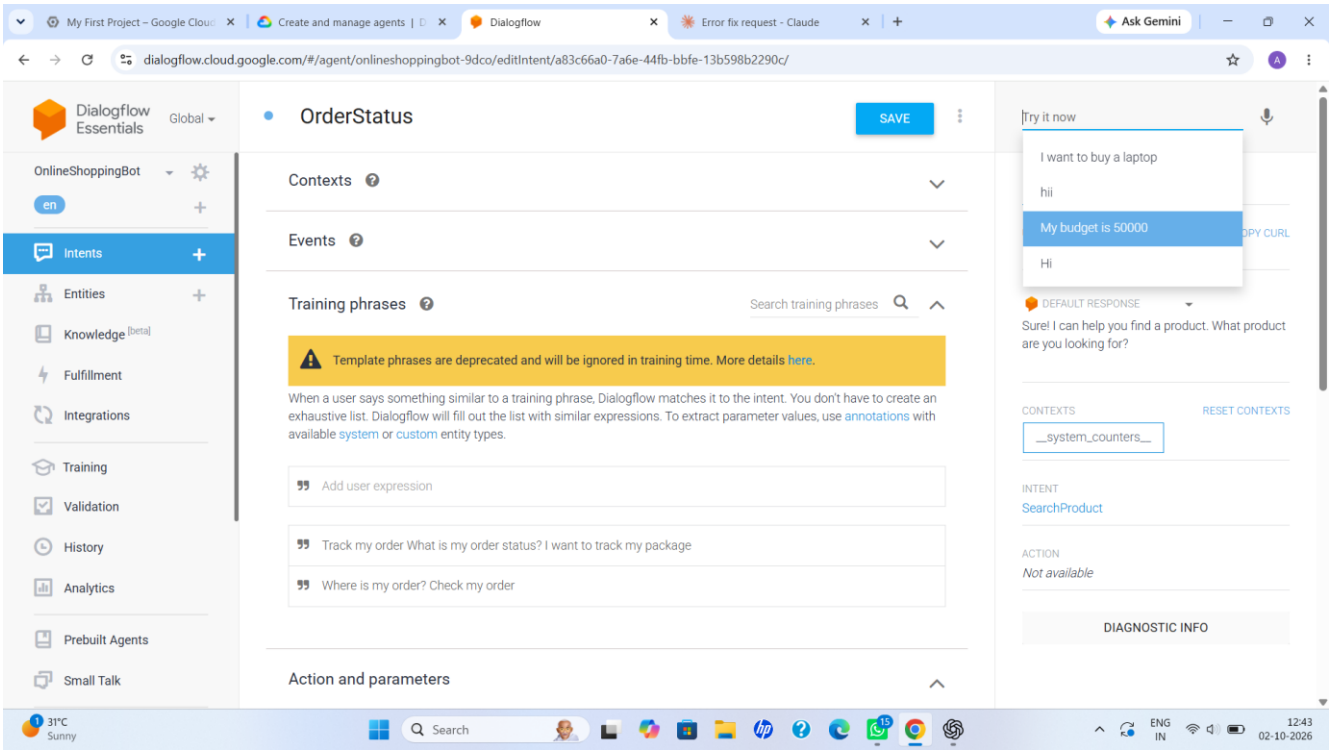
- Understanding of Google Cloud project creation and cloud service usage.
- Understanding of Dialogflow ES agents and conversational flows.
- Understanding of intents and training phrases.
- Understanding of entities and parameter extraction.
- Experience in testing a chatbot using real user queries.
- Understanding of how a chatbot can be integrated with serverless and database services.

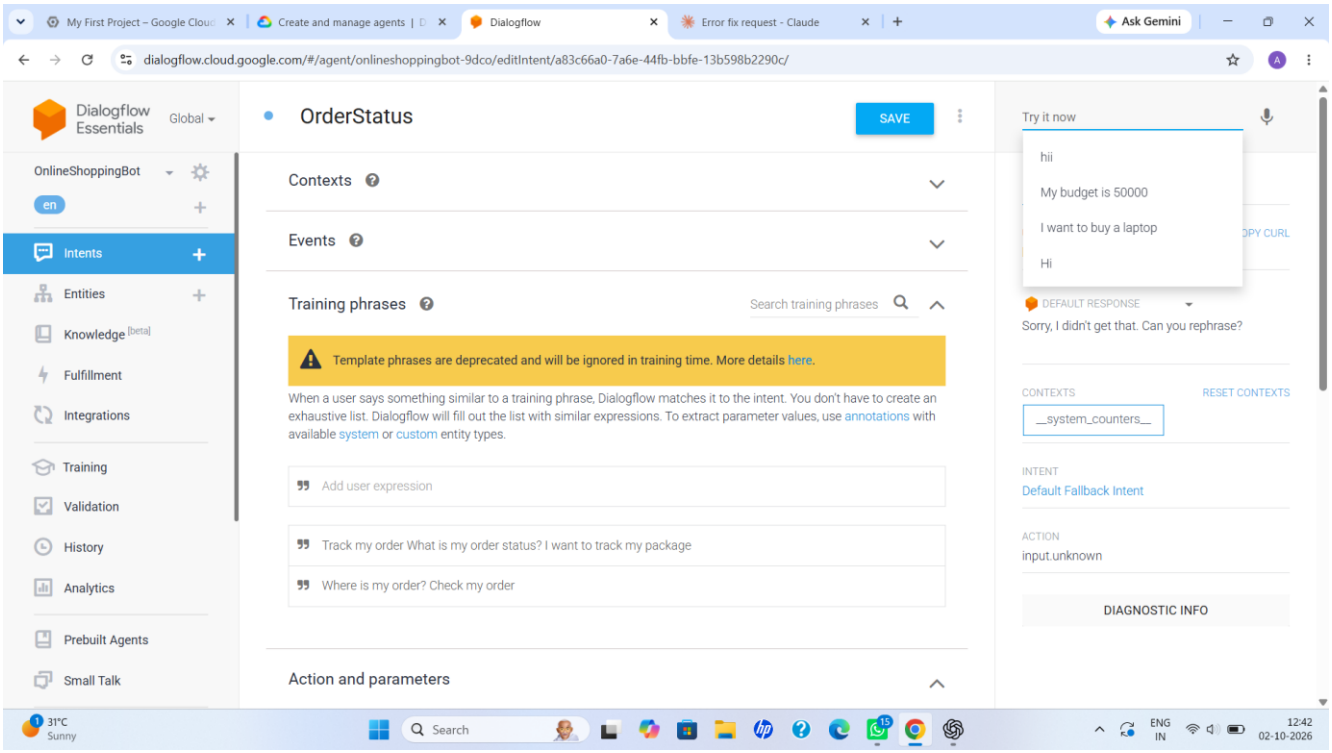
19. References

1. Google Cloud Dialogflow ES Documentation

2. Google Cloud Platform Documentation

3. Dialogflow ES Console and project configuration used during implementation


My First Project – Google Cloud
Create and manage agents | D
Dialogflow
Error fix request - Claude
Ask Gemini
dialogflow.cloud.google.com/#/agent/onlineshoppingbot-9dco/editIntent/a83c66a0-7a6e-44fb-bbfe-13b598b2290c/
Dialogflow Essentials
Global
OnlineShoppingBot
en
Intents
Entities
Knowledge [beta]
Fulfillment
Integrations
Training
Validation
History
Analytics
Prebuilt Agents
Small Talk
OrderStatus
SAVE
Contexts
Events
Training phrases
Search training phrases
Template phrases are deprecated and will be ignored in training time. More details here.
When a user says something similar to a training phrase, Dialogflow matches it to the intent. You don't have to create an exhaustive list. Dialogflow will fill out the list with similar expressions. To extract parameter values, use annotations with available system or custom entity types.
Add user expression
Track my order What is my order status? I want to track my package
Where is my order? Check my order
Action and parameters
Try it now
I want to buy a laptop
hii
My budget is 50000
Hi
DEFAULT RESPONSE
Sure! I can help you find a product. What product are you looking for?
CONTEXTS
RESET CONTEXTS
__system_counters__
INTENT
SearchProduct
ACTION
Not available
DIAGNOSTIC INFO
31°C Sunny
Search
ENG IN
12:43
02-10-2026


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dialogflow.cloud.google.com/#/agent/onlineshoppingbot-9dco/editIntent/a83c66a0-7a6e-44fb-bbfe-13b598b2290c/
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en
Intents
Entities
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I want to buy a laptop
Hi
DEFAULT RESPONSE
Sorry, I didn't get that. Can you rephrase?
CONTEXTS
RESET CONTEXTS
__system_counters__
INTENT
Default Fallback Intent
ACTION
input.unknown
DIAGNOSTIC INFO
31°C Sunny
Search
ENG IN
12:42
02-10-2026